

Assignment 2

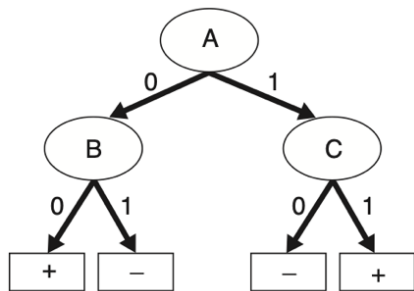
Due: March 16, 2026 at 11:59 pm (Central Time)

Q1. Consider training examples shown in the following table for a binary classification problem.

- i. (10%) Compute the Gini index for the Gender attribute.
- ii. (10%) Compute the Gini index for the Car Type attribute using multiway split.
- iii. (10%) Compute the Gini index for the Shirt Size attribute using multiway split.
- iv. (10%) Which attribute is better, Gender, Car Type, or Shirt Size? Why?

Gender	Car Type	Shirt Size	Class
M	Family	Small	C0
M	Sports	Medium	C0
M	Sports	Medium	C0
M	Sports	Large	C0
M	Sports	Extra Large	C0
M	Sports	Extra Large	C0
F	Sports	Small	C0
F	Sports	Small	C0
F	Sports	Medium	C0
F	Luxury	Large	C0
M	Family	Large	C1
M	Family	Extra Large	C1
M	Family	Medium	C1
M	Luxury	Extra Large	C1
F	Luxury	Small	C1
F	Luxury	Small	C1
F	Luxury	Medium	C1
F	Luxury	Medium	C1
F	Luxury	Medium	C1
F	Luxury	Large	C1

Q2. Consider the following decision tree classifier (on the left side) and test set (on the right side):



ID	A	B	C	Label
1	0	1	0	+
2	0	1	1	+
3	0	0	0	+
4	0	0	1	-
5	1	0	0	-
6	1	0	1	-
7	1	1	0	-
8	1	1	1	-

- (10%) Use this decision tree to classify each data point in the given test set and provide the predicted class for each data point.
- (10%) Compute the confusion matrix (i.e., TP, FP, TN, FN) based on the decision tree's predictions on the test set.
- (10%) Compute the performance of this decision tree on the given test set in terms of Accuracy, Precision, Recall, and F1.

Q3. You are given a *training dataset* called “Iris Dataset.” It contains four features (length and width of sepals and petals) of 12 samples of three species of Iris (Iris setosa, Iris virginica and Iris versicolor). The training dataset is shown in the Figure (A). The column “variety” describes the species of each Iris example. Given this training dataset, we want to use *kNN classification algorithm* to classify three unknown test examples (in the Figure (B)) into their correct species (Iris setosa, Iris virginica or Iris versicolor). The *distance* between each test example to each of the 12 training examples is provided in Figure (C). Given these pieces of information, please answer the following questions:

- (15%) If $k=1$, what is the prediction for each test example and why?
- (15%) If $k=3$, what is the prediction for each test example and why?

(A) Training Dataset

	sepal.length	sepal.width	petal.length	petal.width	variety
Example					
0	4.6	3.2	1.4	0.2	Setosa
1	6.7	3.1	4.4	1.4	Versicolor
2	6.3	3.4	5.6	2.4	Virginica
3	6.4	2.9	4.3	1.3	Versicolor
4	5.0	2.0	3.5	1.0	Versicolor
5	4.6	3.6	1.0	0.2	Setosa
6	5.8	2.7	5.1	1.9	Virginica
7	4.8	3.0	1.4	0.3	Setosa
8	7.1	3.0	5.9	2.1	Virginica
9	6.3	2.9	5.6	1.8	Virginica
10	5.7	4.4	1.5	0.4	Setosa
11	6.9	3.1	4.9	1.5	Versicolor

(B) Test Dataset

	sepal.length	sepal.width	petal.length	petal.width
Test Example				
0	4.6	3.4	1.4	0.3
1	5.7	2.9	4.2	1.3
2	5.9	3.0	5.1	1.8

(C) Distance from Test Examples to Train Examples

	Distance to Example 0	Distance to Example 1	Distance to Example 2	Distance to Example 3	Distance to Example 4	Distance to Example 5	Distance to Example 6	Distance to Example 7	Distance to Example 8	Distance to Example 9	Distance to Example 10	Distance to Example 11
Test Example												
0	0.2	3.8	5.0	3.6	2.6	0.5	4.3	0.4	5.5	4.8	1.5	4.4
1	3.2	1.0	1.9	0.7	1.4	3.6	1.1	3.1	2.3	1.6	3.2	1.4
2	4.2	1.1	1.0	1.1	2.2	4.6	0.3	4.1	1.5	0.6	4.1	1.1